

13 (a)

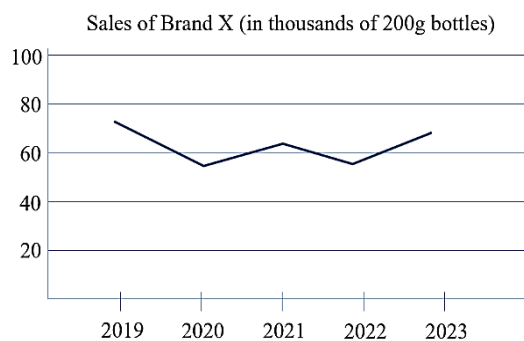
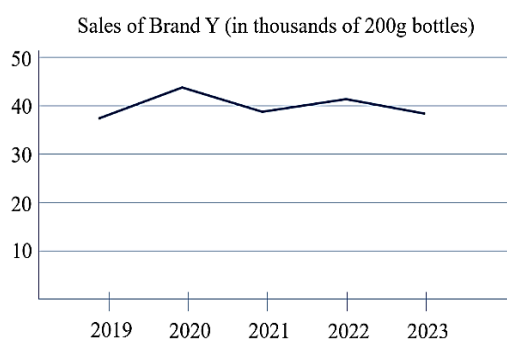


Two bottles of Brand A coffee are geometrically similar.  
The larger bottle contains 200 g of coffee granules.

The larger bottle is approximately 49% taller than the smaller bottle.  
Find, in grams, the amount of coffee granules in the smaller bottle.  
Give your answer to the nearest gram.

Answer \_\_\_\_\_ g [2]

(b)



Brand Y claims that its sales of 200 g bottles of coffee are better than that of Brand X in the past 5 years.  
State one aspect of the charts that makes this claim wrong.

Answer \_\_\_\_\_

[1]

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- 14 (a) Express 1260 as a product of its prime factors.

Answer \_\_\_\_\_ [1]

- (b) The highest common factor of two numbers is 12 while the lowest common multiple is 1260. Both numbers are less than 200.  
Find the two numbers.

Answer \_\_\_\_\_ and \_\_\_\_\_ [2]

- (c)  $486 = 2 \times 3^5$ .  
 $m$  and  $n$  are both prime numbers.

Find the values of  $m$  and  $n$  so that  $486 \times \frac{m}{n}$  is both a perfect square and a perfect cube.

Answer  $m =$  \_\_\_\_\_ and  $n =$  \_\_\_\_\_ [1]

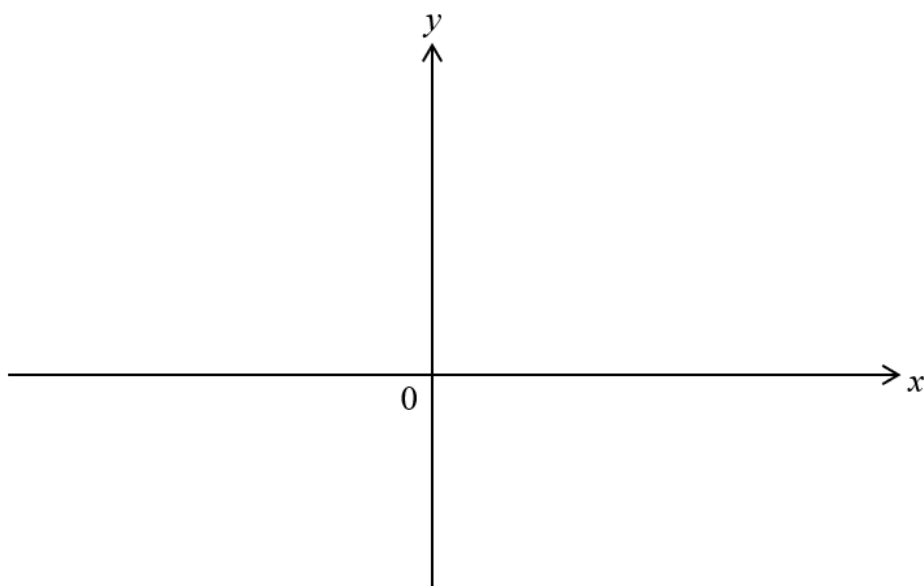
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- 15 (a) Express  $x^2 + \frac{1}{2}x + 1$  in the form  $(x + a)^2 + b$ .

Answer \_\_\_\_\_ [2]

- (b) Sketch the graph of  $y = x^2 + \frac{1}{2}x + 1$  on the axes below.  
Indicate clearly the coordinates of the point where the graph crosses the y-axis  
and the turning point on the curve.

Answer



[2]